# MATH 3330 Homework 9

Siefert-van Kampen Theorem

Suppose $X$ is a space which is the union of closed subspaces $Y$ and $Z$ and there is a point $p$ such that $Y \cap Z = \{p\}$. Then we call $X$ the wedge of $Y$ and $Z$ and write $X = Y \vee Z$.

Question 1

Find $\pi_1(S^1 \vee S^1)$. Explain your reasoning in terms of the Siefert-van Kampen Theorem.

Using the Siefert-van Kampen Theorem, we can decompose $S^1 \vee S^1$ into a two pieces that are glued together at a single point. So let $Y = S^1$ be the first loop and $Z = S^1$ be the second loop which are joined at a point $p$ where $Y \cap Z = \{p\}$. Since $Y \cap Z$ is contractible to $p$, then $\pi_1(Y \cup Z) \cong \pi_1(Y) * \pi_1(Z)$ where $*$ is the free product. Since $Y$ and $Z$ are both just $S^1$, both of $\pi_1(Y), \pi_1(Z) \cong \mathbb{Z}$. Then $\pi_1(S^1 \vee S^1) = \mathbb{Z} * \mathbb{Z}$.

Being in the free group here means that order matters. Wrapping around the $Y$ loop then the $Z$ loop is different than doing that order backwards.

Question 2

Assume $p$ is the deformation retract of of some $W_1$ in $Y$ and of some $W_2$ in $Z$. What is $\pi_1(Y \vee Z)$. Explain your reasoning in terms of the Seifert-van Kampen Theorem.

$\pi_1(Y \vee Z) = \pi_1(Y) * \pi_1(Z)$. (* is the free product)

We can let $U$ be an open neighborhood of $Y$ in $Y \vee Z$ that deformation retracts into $Y$, and $V$ be an open neighborhood of $Z$ in $Y \vee Z$ that deformation retracts into $Z$. Then $U \cup V = Y \vee Z$ and $U \cup V$ is an open neighborhood of $p$ that deformation retracts to that point. Then since $\pi_1(U \cap V)$ is isomorphic to the identity, $\pi_1(Y \vee Z) \cong \pi_1(Y) *_{\pi_1(U \cap V)} \pi_1(Z) = \pi_1(Y) * \pi_1(Z)$ where $*$ is the free product.

Question 3

Explain how to find the fundamental group of both the standard torus and the real projective plane using the Seifert-van Kampen Theorem.

**Torus:**
$T = S^1 \times S^1$. Let $U = T \setminus p$ for a point $p$ on the torus, and $V$ be a small open ball around $p$.

1. $U \cap V$ retracts to $S^1$ which is the same as $\mathbb{Z}$.
2. $V$ can be contracted to a point and is trivial.
3. $U$ contracts to the frame which is $aba^{-1}b^{-1}$ and is the same as $\mathbb{Z} * \mathbb{Z}$.

Then $\pi_1(T) = (\mathbb{Z} * \mathbb{Z}) *_{\mathbb{Z}} e$. Figuring out the amalgamation:
$\pi_1(T) = \langle a, b \mid aba^{-1}b^{-1} = e \rangle = \langle a, b \mid ab = ba \rangle = \mathbb{Z} \times Z$ which is the abelian group on two generators.

**Real Projective Plane:**

1. $U = \mathbb{R}P^2 \setminus p$ for some point $p$. This retracts to the frame, and is then the same as $\mathbb{Z}$
2. $V$ is a disk around the point $p$ which retracts to a trivial point.
3. $U \cap V$ retracts to $S^1$ which is the same as $\mathbb{Z}$.

Then $\pi_1(\mathbb{R}P^2) \cong \mathbb{Z} *_{\mathbb{Z}} e$. Figuring out the amalgamation: $\pi(\mathbb{R}P^2) = \langle 1, a \mid i_U(\gamma) = i_V(\gamma) \rangle$. $i_V(\gamma) = e$ since loops in $V$ are trivial. $i_U(\gamma) = a^2$ since loops in $U$ must make two loops around $p$ in order to return to their starting point. So, $\pi_1(\mathbb{R}P^2) = \langle 1, a \mid b^2 = e \rangle = \mathbb{Z} \setminus 2\mathbb{Z}$ which is the cyclic group of order 2.

> Question 4
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> Let's find the fundamental group of the 2-fold torus two ways.
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> 1. Find the fundamental group of the 2-fold torus using the appropriate identification of a polygon. Explain using Sefiert-van Kampen( - are you noticing a theme?).
> 2. Find the fundamental group of the 2-fold torus by viewing it as the connect sum of two standard tori and use Seifert-van Kampen where U and V are the punctured tori and the intersection is a cylinder.

4. We can represent the standard 2-fold torus (call it $X$) as $aba^{-1}b^{-1}cdc^{-1}d^{-1}$ (by gluing two standard tori together) in the shape of an octagon. Let $U$ be that octagon with a single point $p$ missing and let $V$ be a small disk around that point. Then $U \cap V$ is homotopy equivalent to $S^1$. $U$ retracts to the boundary, and is homotopy equivalent to the wedge of circles (one for each edge $a, b, c, d$). $V$ contracts to a point and is trivial. Then $\pi_1(X) = \langle a, b, c, d \mid aba^{-1}b^{-1}cdc^{-1}d^{-1} = e \rangle$ is the group on four generators.
5. Let $U$ and $V$ be two tori with a disk removed and $U \cap V$ be a cylinder that connects them. $\pi_1(U) = \langle a, b \rangle$ the free group on two generators, and $\pi_1(V) = \langle c, d \rangle$ the free group on two generators. $\pi_1(U \cap V) \cong \mathbb{Z}$. Then, $\pi_1(T_1 \# T_2) = \langle a, b \rangle *_{\mathbb{Z}} \langle c, d \rangle$. When a loop crosses the boundary between $T_1$ (with the polygon representation $aba^{-1}b^{-1}$) and $T_2$ (with the polygon representation $cdc^{-1}d^{-1}$), these must be equal $aba^{-1}b^{-1} = cdc^{-1}d^{-1}$. So $aba^{-1}b^{-1}dcd^{-1}c^{-1} = e$. $c$ and $d$ can be relabeled with each other, so

$aba^{-1}b^{-1}cdc^{-1}d^{-1} = e$. Then the fundamental group is $\langle a, b, c, d \mid aba^{-1}b^{-1}cdc^{-1}d^{-1} = e \rangle$ the group on four generators.

### Question 5

Find spaces $X$ and $Y$ with $\pi_1(X, p) \cong \mathbb{Z}_n \times \mathbb{Z}_m$ and $\pi_1(Y, p) \cong \mathbb{Z}_n * \mathbb{Z}_m$

$X = L(n, 1) \times L(m, 1)$ is the product of two lens spaces. $L(n, 1) \cong \mathbb{Z}_n$ so then $\pi_1(X) = \pi_1(L(n, 1) \times L(m, 1)) = \pi_1(L(n, 1)) \times \pi_1(L(m, 1)) = \mathbb{Z}_n \times \mathbb{Z}_m$.

$Y = L(n, 1) \vee L(m, 1)$. By the Siefert-van Kampen Theorem, the wedge of two spaces is isomorphic to their free product: $\pi_1(L(n, 1) \vee L(m, 1)) \cong \pi_1(L(n, 1)) * \pi_1(L(m, 1)) \cong \mathbb{Z}_n * \mathbb{Z}_m$.

### Question 6

(Not on Test 5 but could be on final exam) For each of the following surface presentations, compute the Euler characteristic and determine which standard surface it represents.

1. $S = \left\langle a, b, c, d \mid abcdad^{-1}cb^{-1} \right\rangle$.
2. $S = \left\langle a, b, c, d, e, f, g, h, i, j \mid abfg;\ bchi;\ cdjf^{-1};\ deg^{-1}h^{-1};\ eai^{-1}j^{-1} \right\rangle$.

1. Each edge appears twice, so $abcdad^{-1}cb^{-1}$ is a proper labeling scheme of projective plane type (since there are letters of like-powers). There is 1 face, 4 edges, and 2 vertices (the tails of $b$ and the heads of $d$ are the same, and both sides of the $c$ edges are the same), so $1 - 4 + 2 = 2 - m \to m = 3$. This is a 3-fold projective plane.
2. We can concatenate identity loops where they have shared edges:

$$\begin{aligned}
&abfg \\
&a[bchi]bfg \\
&ab[cdjf^{-1}]chibgf \\
&abc[deg^{-1}h^{-1}]djf^{-1}chibgf \\
&abcd[eai^{-1}j^{-1}]eg^{-1}h^{-1}djf^{-1}chibgf \\
&abcdeai^{-1}j^{-1}eg^{-1}h^{-1}djf^{-1}chibgf
\end{aligned}$$

   Then since each letter appears twice and there are letters with like-powers, this is a projective plane. There is 1 face, 10 edges, and 4 vertices, so $1 - 10 + 4 = 2 - m \to m = 7$. This is a 7-fold projective plane.